

# Suricata as IPS (Intrusion Prevention System)

When Suricata runs as an **IDS**, it only detects and logs attacks.

Attack

Suricata IDS

Alert Generated

When Suricata runs as an **IPS**, it can detect **and block** malicious traffic.

Attack

Suricata IPS

Block Packet

---

## Lab Topology

Kali Linux  
192.168.1.10

Suricata IPS  
192.168.1.20

Victim  
192.168.1.30

Suricata sits inline between attacker and victim.

---

## Step 1: Install Suricata

Ubuntu:

```
sudo apt update  
  
sudo apt install suricata -y
```

Verify:

```
suricata -V
```

---

## Step 2: Enable NFQUEUE Support

Check:

```
suricata --build-info | grep NFQ
```

Expected:

```
NFQ support: yes
```

NFQUEUE is used by iptables to send packets to Suricata.

---

## Step 3: Create IPS Rule

Open:

```
sudo nano /etc/suricata/rules/local.rules
```

Add:

```
drop icmp any any -> any any(msg:"PING BLOCKED";sid:1000001;rev:1;)
```

Meaning:

```
If ICMP packet detected  
Drop packet
```

---

## Step 4: Enable Rule

Edit:

```
sudo nano /etc/suricata/suricata.yaml
```

Ensure:

```
rule-files:  
- local.rules
```

---

## Step 5: Test Configuration

```
sudo suricata -T -c /etc/suricata/suricata.yaml
```

Expected:

```
Configuration provided was successfully loaded
```

---

## Step 6: Configure iptables

Send packets to NFQUEUE:

```
sudo iptables -I INPUT -j NFQUEUE
```

Verify:

```
sudo iptables -L -n
```

Expected:

```
NFQUEUE
```

---

## Step 7: Start Suricata in IPS Mode

```
sudo suricata -c /etc/suricata/suricata.yaml -q 0
```

Where:

```
-q 0
```

means:

```
Use NFQUEUE 0
```

---

## Step 8: Test Blocking

From Kali:

```
ping 192.168.1.20
```

Expected:

```
Request timed out
```

Because:

Suricata

Matched Rule

Dropped Packet

---

## Block SSH

Rule:

```
drop tcp any any -> any 22  
(msg:"SSH BLOCKED";  
sid:1000002;  
rev:1;)
```

Test:

```
ssh user@192.168.1.20
```

Connection blocked.

---

## Block Telnet

Rule:

```
drop tcp any any -> any 23  
(msg:"TELNET BLOCKED";  
sid:1000003;  
rev:1;)
```

---

# Block SQL Injection

Rule:

```
drop http any any -> any any
(msg:"SQL INJECTION BLOCKED";
content:"UNION SELECT";
nocase;
sid:1000004;
rev:1;)
```

Test:

```
curl "http://victim/test.php?id=1 UNION SELECT 1,2"
```

Packet dropped.

---

## View Alerts

Fast log:

```
sudo tail -f /var/log/suricata/fast.log
```

Example:

```
PING BLOCKED
```

JSON log:

```
cat /var/log/suricata/eve.json
```

---

## Remove IPS Mode

Remove iptables rule:

```
sudo iptables -F
```

Stop Suricata:

```
sudo systemctl stop suricata
```

---

## Packet Flow

Incoming Packet

NFQUEUE

Suricata

Rule Match?

No    Yes

Forward   Drop

---

## IDS vs IPS in Suricata

| IDS           | IPS            |
|---------------|----------------|
| alert rule    | drop rule      |
| Detect only   | Detect + Block |
| Packet passes | Packet dropped |

|                 |             |
|-----------------|-------------|
| Monitoring mode | Inline mode |
|-----------------|-------------|

Example IDS rule:

```
alert icmp any any -> any any
(msg:"Ping Detected";
sid:1;)
```

Example IPS rule:

```
drop icmp any any -> any any
(msg:"Ping Blocked";
sid:2;)
```

---

### How does Suricata work as an IPS?

Suricata works as an IPS by receiving packets through **NFQUEUE**, inspecting them against rules, and using **drop** actions to block malicious traffic before it reaches the destination.